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HGC³AE² at the Degraded Edge

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Abstract

HGC³AE² is the load-bearing governance framework introduced in Paper 1 of the AI Governance series; this paper applies it to the degraded edge. The framework is link-agnostic in shape and link-aware in mechanism: the same governor structure that governs agentic ops at the data center governs distributed state at the C5ISR edge. The H half governs human authority and scope; the C³ half curates content, catalog, and curriculum into a doctrine that survives the link; the AE² half authenticates identity and enforces the boundaries that make autonomy safe. This paper — Paper 2 of *The Implications of Edge Degraded Ops* — examines what each half does when the link is marginal or absent and specifies the §6 governor-application contract that every subsequent paper in the series anchors. Where the survey-frame paper (Tactical Substrate) describes the architecture at width, this paper depths the governance dimension.

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ABSTRACT

The HGC³AE² framework specifies the structural commitments — human-governed authority, curated context, agentially engineered execution, runtime verification — that distinguish trustworthy operation from capable but ungoverned operation in agentic systems.¹ Those commitments were established in conditions that the cloud-native deployment substrate satisfies by default: continuous connectivity to authoritative governance, accessible curated-context tiers, observability planes that aggregate continuously, and adversarial pressure assumed to be commercial rather than operational. The Edge Degraded Operations substrate (EDO-P1) satisfies none of these by default.² This paper develops HGC³AE² as it must operate under DDIL conditions: how Human-governance is preserved through pre-authorized envelopes that survive disconnection; how Curated Context survives partition; how Cybersecurity is enforced under adversarial pressure on every layer; how Continuity becomes the architecture's primary design property; how Coherence is maintained across reconciliation cycles; how Agentic Engineering operates within disconnection-tolerant envelopes; and how Execution is verified runtime when the verification machinery itself may be partially disconnected. The argument is that HGC³AE² is preserved under DDIL conditions, not weakened — the architectural commitments survive translation to the tactical substrate, provided the translation is done explicitly rather than assumed.

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1. INTRODUCTION

The HGC³AE² framework was developed against the cloud-native deployment substrate: continuous connectivity, accessible centralized governance, observable persistence, commercial adversarial pressure.³ The framework's structural commitments — what humans must do, what curation must preserve, what agentic execution must satisfy, how runtime verification closes the loop — were specified in terms of operational primitives that the cloud-native substrate supports natively.

The Edge Degraded Operations substrate (EDO-P1) supports those primitives differently. Connectivity is intermittent; centralized governance may be unreachable; persistence is anchored at the edge node with reconciliation across boundaries; adversarial pressure is operational rather than commercial.⁴ HGC³AE²'s commitments do not become irrelevant under these conditions; they become **load-bearing in a different way**. The H of HGC³AE² cannot rely

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).

3 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

4 EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).



on continuous human consultation; it must rely on pre-authorized envelopes that survive disconnection. The G of curated context cannot rely on a centralized tier; it must operate over local curated state with explicit reconciliation discipline. The C¹ of cybersecurity must assume hostile substrate rather than benign substrate.

This paper translates each letter of HGC³AE² into its degraded-edge form. The argument is that the framework's commitments are preserved under DDIL conditions; the translation is the architectural work that the Edge Degraded Operations Undecalogy carries out across its eleven papers. §2 names five failure modes that occur when cloud-native HGC³AE² is deployed onto the tactical substrate without translation. §3 specifies the letter-by-letter translation. §4 treats the translation as a continuous process. §5 binds the translated framework to Skipjack. §6 returns the analysis to HGC³AE² and frames the remainder of the Undecalogy.

2. FAILURE MODES OF UNTRANSLATED HGC³AE² ON THE TACTICAL SUBSTRATE

Five failure modes recur when cloud-native HGC³AE² implementations are deployed onto the tactical substrate without explicit translation.

Halt-on-disconnect. A cloud-native implementation of H — continuous human governance consultation — halts when governance is unreachable. The agentic system loses operational capability, often precisely when operational capability is most needed.⁵

Stale-curation drift. Local curated context drifts from authoritative state during disconnection. When connectivity returns, reconciliation surfaces divergence the system has been operating against. Without explicit drift handling, the divergence either accumulates silently or forces costly rollback.⁶

Privileged-channel collapse. Cloud-native security assumes privileged channels (KMS, IAM, attestation services) are continuously reachable. Under disconnection, these channels collapse; security state cannot be verified; the system either operates against stale verification (risky) or halts (also risky).⁷

Observability blackout. Continuous observability assumes continuous connectivity to the centralized telemetry plane. Under disconnection, telemetry buffers locally; when capacity is constrained, buffers exhaust; observability data is lost. Operator awareness of the system's state during disconnection becomes a reconstructed estimate at best.⁸

Reconciliation gaps. Reconnection does not automatically merge the disconnected substrate's state with the authoritative source. Without explicit reconciliation discipline, the substrate either over-writes authoritative state (loss), is over-written by it (loss), or operates in inconsistent dual-state (drift).

5 Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007.

6 Marc Shapiro et al., "Conflict-Free Replicated Data Types," SSS '11, 2011, https://doi.org/10.1007/978-3-642-24550-3_29.

7 Ali Humayed et al., "Cyber-Physical Systems Security — A Survey," *IEEE Internet of Things Journal* 4, no. 6 (2017): 1802–1831, <https://doi.org/10.1109/JIOT.2017.2703172>.

8 Saksham Tushar et al., "Observability in Microservices: A Systematic Literature Review," *Information and Software Technology* 158 (2023): 107173, <https://doi.org/10.1016/j.infsof.2023.107173>.



The unified problem: **cloud-native HGC³AE² implementations fail structurally on the tactical substrate**. The remainder of this paper specifies the translation that preserves the framework.

3. HGC³AE² LETTER-BY-LETTER AT THE DEGRADED EDGE

3.1 H — HUMAN-GOVERNED VIA PRE-AUTHORIZED ENVELOPES

Human governance under DDIL is exercised through **pre-authorized envelopes**: explicit, persistent specifications of what the agentic system may do, against what context, with what authority, while disconnected from continuous human consultation. The envelope is authored by the governance layer when connectivity is available; it persists locally on the substrate; the substrate operates inside the envelope without further authorization until reconnection.⁹

The envelope is not a static permission set. It is a structured specification with explicit conditions for proceeding, conditions for halting, conditions for escalation, and conditions for fallback. The envelope encodes what humans authorized; the substrate enforces the encoding.

3.2 G — CURATED CONTEXT UNDER PARTITION

Local curated context survives partition by being explicitly local-authoritative within the substrate during disconnection, with explicit reconciliation discipline upon reconnection. The CRDT primitives developed in EDO-P4 are the operational instruments: distributed-state types that admit concurrent updates with deterministic merge semantics.¹⁰¹¹ Curation discipline is preserved through local-authoritative status rather than through continuous central authority.

3.3 C¹ — CYBERSECURITY UNDER HOSTILE SUBSTRATE

Security under the degraded edge assumes hostile substrate: cryptographic privacy, integrity attestation, and signed governance events are first-class concerns at every layer. Privileged-channel design must include local-fallback modes for the disconnected case: local key material with explicit rotation discipline; local attestation chains with reconciliation-time verification; signed governance event logs that the substrate can verify offline.¹²¹³

3.4 C² — CONTINUITY AS PRIMARY DESIGN PROPERTY

Continuity becomes the architecture's primary design property under DDIL. Workloads survive disconnection by operating against pre-authorized envelopes; persistence survives connectivity loss by anchoring on local substrate with explicit reconciliation; governance state

9 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

10 Marc Shapiro et al., "Conflict-Free Replicated Data Types," SSS '11, 2011, https://doi.org/10.1007/978-3-642-24550-3_29.

11 EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

12 Bob Callaway et al., "Sigstore: Software Signing for Everybody," CCS '22, 2022, <https://doi.org/10.1145/3548606.3560596>.

13 Jingbin Liu, Yu Qin, Dengguo Feng, "RIM-EI: Toward Trusted Attestation of Persistent State in Trusted Execution Environments," *IEEE Transactions on Dependable and Secure Computing* 19, no. 4 (2022): 2580–2596, <https://doi.org/10.1109/TDSC.2021.3060127>.



survives partition by carrying forward the most recently authoritative envelope until reconciliation updates it.

3.5 C³ — COHERENCE ACROSS RECONCILIATION CYCLES

Coherence is maintained not by continuous availability but by **reconciliation cycles**. Each cycle merges the substrate's accumulated state with the authoritative source's state using deterministic merge semantics. Coherence is therefore episodic-but-correct, not continuous; the architectural commitment is that the reconciliation merge is deterministic and verifiable.¹⁴

3.6 A — AGENTICALLY ENGINEERED WITHIN DISCONNECTION-TOLERANT ENVELOPES

Agentic execution operates within disconnection-tolerant envelopes specified by the governance layer at pre-authorization time. The agentic system does not extend its envelope under disconnection; it operates inside it. Capability without permission is the cloud-native failure mode HGC³AE² names; capability without reachability is the DDIL variation, and the architectural response is the same — the envelope, pre-authorized and locally enforced.

3.7 E — RUNTIME VERIFICATION UNDER PARTIAL DISCONNECTION

Runtime verification under DDIL operates locally: the substrate verifies its own runtime against pre-authorized invariants without continuous external consultation. Verification machinery must itself survive disconnection; the substrate must be able to verify its operation even when external verification services are unreachable.

3.8 THE EXPONENT — THE DISCONNECTED-RECONCILE-ADJUST LOOP

The HGC³AE² exponent closes through the disconnected-reconcile-adjust cycle (EDO-P1 §4). Telemetry accumulated during disconnection feeds the next reconciliation; reconciliation feeds governance-layer envelope adjustment; the adjusted envelope propagates to the substrate for the next disconnected operating cycle. The loop closes asynchronously; the closure is preserved.

4. THE TRANSLATION AS A CONTINUOUS PROCESS

The letter-by-letter translation is not a one-time architectural exercise. The substrate's operating conditions evolve: connectivity profiles change, adversarial pressure intensifies, capacity oscillations shift. The translation must be revisited at sprint-aligned Skipjack cadence, with envelope adjustments propagating the revised translation to the substrate.

The three-phase cycle from EDO-P1 (disconnected operation → reconciliation → adjustment) is the operational instantiation of the translation's continuity.

¹⁴ Werner Vogels, "Eventually Consistent," *Communications of the ACM* 52, no. 1 (2009): 40–44, <https://doi.org/10.1145/1435417.1435432>.



5. SKIPJACK COUPLING UNDER TRANSLATION

The Skipjack mechanisms operate under the translated HGC³AE² with the adaptations introduced in EDO-P1 §5.¹⁵¹⁶ **Context integrity** holds locally during disconnection and is verified during reconciliation. **Structured routing** operates within the pre-authorized envelope. **HITL control points** fire when envelope boundaries are approached; under disconnection, control points become *halt-and-await-reconnection* events. **Iterative feedback** closes asynchronously across reconciliation cycles.

The mechanisms preserve their commitments under translation. The Skipjack Protocol is one of the architectural commitments HGC³AE² depends on, and its translation to the tactical substrate is what makes the framework operational at the degraded edge.

6. IMPLICATIONS AND CONCLUSION

The implications of translating HGC³AE² to the degraded edge extend to architecture (pre-authorized envelopes become first-class artifacts), tooling (CRDT and WAL primitives, attestation chains, local verification machinery require dedicated investment), operational doctrine (DDIL ops is a discipline), and adoption sequencing (organizations move to the degraded edge in phases, not all at once).

The core claim: **HGC³AE²'s structural commitments are preserved under DDIL conditions through explicit translation; the translation is the architectural work that the Edge Degraded Operations Undecalogy carries out across its eleven papers.**

Returned to HGC³AE² letter by letter — this paper's §3 is the return; the rest of the Undecalogy develops each translated letter in operational depth.¹⁷¹⁸¹⁹²⁰

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15 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

16 EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).

17 EDO-P3 *When the Link Is the Variable* promotion (in concurrent drafting; yks-build#95).

18 EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

19 EDO-P5 *Orchestration Problem* promotion (in concurrent drafting; yks-build#97).

20 EDO-P11 *Deployment Problem* CAPSTONE promotion (in concurrent drafting; yks-build#110).